# Diffusion of Hydrogen in Iron

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A mass spectrometer was used to study hydrogen diffusion and trapping phenomena in fully-annealed and slightly cold-worked pure iron specimens which were in contact with distilled water or dilute acidic Buffer solutions. In the case of fully-annealed iron and slightly cold-worked iron, hydrogen can diffuse into iron only when the iron contacts water directly. This diffusion phenomenon of hydrogen increased markedly with temperature and was accelerated by abrasion and hydrogen ion concentration in dilute acid. Abrasion and hydrogen ions in a dilute acidic Buffer solution did not affect the diffusion coefficient of hydrogen, $D$, but increased the hydrogen concentration at the iron surface contacting water or Buffer solution, $C_S$. The permeability of hydrogen in fully-annealed iron in contact with distilled water and the diffusion coefficient of hydrogen in fully-annealed and slightly cold-worked iron for the temperature range 10° to 100°C were measured. Trapping parameters in the slightly cold-worked iron were calculated.

DIFFUSION of hydrogen plays an important role in the field of stress-corrosion of metals. Norton[1] demonstrated the possibility of diffusion of hydrogen through metal in his use of a steel-shelled ionization gage to measure hydrogen partial pressure. Darken and Smith[2] determined the behavior of hydrogen in steel during and after immersion in acid by measuring the displacement of butyl phthalate in a calibrated burette due to the diffused hydrogen from the acid through and out of a steel disc. They showed the increase of solubility and permeability of hydrogen in SAE 1020 steel with the decrease of pH at 35°C, the increase of hydrogen permeability with temperature and the increase of hydrogen absorption with cold reduction. Frank, Swets, and Fry[3] used a mass spectrometer and an abrasion technique to study diffusion of hydrogen in mild steel in contact with water over the temperature range 25° to 90°C. By taking into consideration the characteristics of the mass spectrometer, they obtained 6.3 kcal per mole for the build-up (stabilization) technique.[1] *

*In the build-up or stabilization technique, the hydrogen concentration builds up from roughly zero to the steady state value.

The same authors[4] considered that clean iron was continually presented to water during abrasion and hydrogen was generated by the chemical reaction between iron and water. Johnson and Hill[5] showed that the cold working of Fe-C alloys increased greatly the hydrogen solubility and decreased the diffusion coefficient at temperatures 200° to 400°C, and that these effects were increasing functions of the deformation degree, and that hydrogen content increased with the carbon content of the specimen. The same authors[6] studied hydrogen diffusion in cylindrical $\alpha$-Fe specimens using hydrogen evolution rate at 25° to 780°C and found that the activation energy above 200°C was 3.2 and 7.8 kcal per mole below 200°C. They postulated that at low temperatures, hydrogen in metals was trapped with an energy of about 5 kcal per mole below that of the interstitial hydrogen. McNabb and Foster[7] developed a modified mathematical equations for lower temperature trapping phenomenon in metals. They assumed that hydrogen was delayed at fixed sites in the lattice and analyzed[8] some measurements of the rate of loss of hydrogen from cylinders of iron and steel in terms of a trapping theory. Nelson[9] obtained an activation energy of 2.8 kcal per mole for a normalized 4130 hollow cylindrical steel specimen from room temperature to 700°C at various pressures of gaseous hydrogen by time-lag techniques using a quadruple residual gas analyzer.

In this study, the suitable mathematical equations for a reasonable fit with the experimental curve will be found using equations derived from the classic Fick's diffusion equation. The diffusion coefficient and activation energy of hydrogen for fully-annealed pure iron in contact with distilled water and dilute acidic Buffer solution will be determined by means of mass spectrometer. The trapping parameters for slightly cold-worked iron specimens will be calculated to study the retardation of the hydrogen diffusion rate.

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Manuscript submitted October 24, 1968.

## MATHEMATICAL BACKGROUND

Let us make the following assumptions for the mathematical treatment: 1) Volume diffusion in the bulk metal is the predominent mechanism for hydrogen diffusion. 2) The saturated hydrogen concentration of the specimen contacting water, $C_S$, and hydrogen diffusion coefficient, $D$, are constant at a given temperature.

Let the vacuum side of the plate specimen be at $x = 0$ and the liquid side be at $x = a$, where $a$ is the thickness of the plate. When the high hydrogen concentration, $C_S$, at the liquid side is maintained, the boundary conditions will be shown as follows:

$t = 0$:

$C = 0$ at $x = 0, x > a$

$C = f(x)$ at $0 < x < a$

$t > 0$:

$C = 0$ at $x = 0, x < 0$, and $x > a$

$C = C_S$ at $x = a$

Then, the general solution[10] of Fick's classic second-order diffusion equation is:

$$C = \frac{C_S x}{a} + \frac{2C_S}{\pi}\sum_{n=1}^{\infty}\frac{\cos n\pi}{n}\sin\left(\frac{n\pi}{a}\right)x\exp\left[-\left(\frac{n\pi}{a}\right)^2 Dt\right]$$

$$+\frac{2}{a}\sum_{n=1}^{\infty}\sin\left(\frac{n\pi}{a}x\right)\exp\left[\left(\frac{n\pi}{a}\right)^2 Dt\right]\int_0^a f(x)\sin\left(\frac{n\pi}{a}x\right)dx \qquad [1]$$

where $f(x)$ is the initial hydrogen concentration in the specimen, usually constant or zero, $t$ is diffusion time, and $n = 1, 2, 3, \cdots\infty$. For simplification, one may take $f(x) = 0$,* since the specimen is kept at a bake-out temperature of 250°C for about 10 days with one side of the specimen exposed to the vacuum system. Therefore, the third term in Eq. [1] will disappear.

*The baking operation will eliminate hydrogen atoms which remain at active traps. Hydrogen atoms captured by traps of deeper energy well than that of active traps cannot be removed by fully-annealing and remain permanently in the material as residual hydrogen and further do not diffuse. If one assumes that deep traps filled with hydrogen atoms behave the same toward hydrogen diffusion as does the normal lattice, then one may consider the initial hydrogen concentration, *f(x)* to be zero. The concepts of the trap will be described in more details elsewhere.

When one side of the specimen contacts water, hydrogen is supplied continually at $x = a$. In the case of the stabilization technique, the hydrogen flux, $J$, at $x = 0$, where the specimen surface is exposed to the ultra high vacuum, is shown to be

$$J = -D\left(\frac{\partial C}{\partial x}\right)_{x=0}$$

$$= -\frac{C_S D}{a} - 2\left(\frac{C_S D}{a}\right)\sum_{n=1}^{\infty}(-1)^n\exp\left[-n^2\frac{t}{\tau}\right] \qquad [2]$$

where

$$\tau = \text{relaxation time} \equiv \frac{a^2}{\pi^2 D} \qquad [3]$$

When the diffusion time is very long, $t\to\infty$, steady state will be obtained, since the second term in Eq. [2] becomes zero:

$$(J)_\infty = -\frac{C_S D}{a} = -\frac{P}{a} \qquad [4]$$

where

$P \equiv C_S D$ = permeability.

From Eq. [4], one can calculate the saturated hydrogen concentration at the surface contacting water, $C_S$, when the diffusion coefficient, $D$, is known, since $(J)_\infty$ can be obtained from the value of the saturated hydrogen flux at steady state. In the two-sided decay technique, in which hydrogen is outgassed at both surfaces, the hydrogen flux at $x = 0$ is

$$J = -2\left(\frac{C_S D}{a}\right)\sum_{n=1}^{\infty}(-1)^n\exp\left[-n^2\frac{t}{\tau}\right] \qquad [5]$$

The hydrogen flux[3] at $x = 0$ in the one-sided decay technique, in which one side of the specimen can be considered to be impermeable to hydrogen, is

$$J = -\frac{4}{\pi}\left(\frac{C_S D}{a}\right)\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{(2n-1)}\exp\left[-\left(\frac{2n-1}{2}\right)^2\frac{t}{\tau}\right] \qquad [6]$$

The diffusion coefficient of hydrogen, $D$, may be calculated from the relaxation time, $\tau$, after finding the best fit between the experimental curve and theoretical curves drawn using Eqs. [2], [5], or [6] in the plot of flux, $J$, vs diffusion time, $t$.

When the hydrogen background in the vacuum system is very low under ultra high vacuum, the mathematical equations above can be used for calculation of the diffusion coefficient, $D$, without considering the gas flow of the mass spectrometer.

When a specimen contains a negligible number of active traps for diffusing hydrogen atoms, *i.e.*, dislocations, vacancies, interstitials, or other defects, Fick's classic diffusion equation is valid. McNabb and Foster[7] (hereafter, abbreviated M-F) developed a trapping theory considering active traps. M-F classified the traps into three families according to the tenacity with which they hold a captured hydrogen atom. 1) Traps with shallow energy wells which have a negligible delaying effect. 2) Traps with deep energy wells which can hold their captives permanently and regard their captives as completely removed from the diffusion over the time intervals and at the temperatures being considered. (Captured hydrogen will be called "residual hydrogen" after fully annealing.) 3) Active traps—all remaining traps are considered active traps.

M-F assumed that one active trap can capture and release only one hydrogen atom at a time and that the active traps can be described in their over-all effect by two parameters $p$ and $k$. M-F postulated that the release probability, $p$, would be dependent on the temperature and nature of the active trap, but independent of the local concentrations of trapped and diffusing hydrogen atoms. $p$ was defined as the mean probability that a member of the active traps containing a hydrogen atom would release it before 1 sec had passed. M-F postulated that the probability that the active traps in a small volume, $\delta V$, capture a hydrogen atom was proportional to $C$, the number of diffusing hydrogen atoms per unit volume, and to $N(1-n)$, the number of vacant traps per unit volume, where $n$ is the fraction occupied. $k$ was defined as the capture parameter such that the number of hydrogen atoms captured per sec in $\delta V(\text{cm}^3)$ would be shown by $kCN(1-n)\delta V$. From the above definition, the capture parameter, $k$, has the unit, cu cm per (sec·trap). From this, the physical meaning of $k$ will be interpreted as the space volume of one unoccupied active trap which can capture one diffusing hydrogen atom before 1 sec has passed. The parameter, $k/p$ (cu cm per trap), will be interpreted as the space volume of one active trap which is capable of capturing or releasing one hydrogen atom. In fact, $k/p$ does show the size of one active trap in a space lattice.

M-F showed the effect of the ratio between the trap density, $N$ (trap per cu cm), and $C_S$ (atom H per cu cm) on diffusion of hydrogen: 1) when $N/C_S \ll 1$, the classic diffusion equation will be applicable, for the trapping phenomenon will be negligible. 2) When $N/C_S \gg 1$, the trapping phenomenon is important.

When hydrogen diffusion with a trapping phenomenon is observed, the size of the active trap, $k/p$ (cu cm per trap), and the trap density, $N$ (traps per cu cm), can be calculated.

The ratio of the quantity* of hydrogen crossing unit

*The quantity of hydrogen crossing unit area at the surface contacting the water, $Q$(STP cc per sq cm), can be obtained from the area of the diagram of $J$(STP cc per sq cm. sec) vs $t$ (sec).

area during decay, $Q_e$ (STP cc per sq cm) and that diffusing during the stabilization stage, $Q_a$ (STP cc per sq cm), for the same value of time,* when the surface concentration of hydrogen is maintained at $C_s$, will be shown by[11]

*It is recommended that one uses the necessary time for the hydrogen flux to reach approximately the background value from the steady state value.

$$\frac{Q_e}{Q_a} \simeq [1 + C_s(k/p)]\left[1 - \frac{2}{\pi} C_s(k/p)\right] \qquad [7]$$

From the equation above, the size of an active trap, $k/p$, can be calculated, when the value of the ratio, $Q_e/Q_a$, is known and $Q_e/Q_a \lesssim 1$.*

*Eq. [7] is not valid for calculating $k/p$, when $Q_e/Q_a > 1$.

The apparent diffusion coefficient in a specimen containing active traps, $D_{\text{app}}$, will be shown by[7]

$$D_{\text{app}} \simeq \frac{1}{1 + N(k/p)} D \qquad [8]$$

where $D$ is the hydrogen diffusion coefficient in fully-annealed iron containing no active traps. From Eq. [8], the trap density, $N$, can be calculated when the value of $k/p$ is known.

The size of an active trap, $k/p$, may be shown by the following equation.[11]

$$\frac{(k/p)}{V_0} = \exp\left(\frac{Q_t}{RT}\right) \qquad [9]$$

where

$V_0 \equiv 4\pi r_0 a_0^2$ = volume of the space lattice containing one hydrogen atom at the interstitial site in a lattice without any active trap, which is considered as a cylinder of base, $\pi a_0^2$, and height, $4r_0$.

$r_0$ = radius of iron atom.

Taking $r_0 \simeq a_0 \simeq 3 \times 10^{-8}$ (cm), then $V_0 \simeq 3.4 \times 10^{-22}$ (cu cm). We may find the ratio, $(k/p)/V_0$, assuming $V_0$ constant over the temperature range 0° to 100°C. The energy well of an active trap, $Q_t$, may be calculated if Eq. [9] is valid.

The author will calculate the apparent diffusion coefficient, $D_{\text{app}}$, using the equations obtained from the classic diffusion equation and then use the trapping theory to calculate the size of an active trap, $k/p$, the trap density, $N$, and the ratio, $(k/p)/V_0$, for slightly cold-worked iron in this study.

## EXPERIMENTAL PROCEDURE

The chemical composition, history of mechanical working and heat treatment, and size of the specimens are shown in Table I. The experimental conditions of abrasion, vacuum system, hydrogen standard leak, and sensitivity of Mass Spectrometer (Consolidated Electrodynamic Corp., Type 21-615) are shown in Table II. The experimental apparatus is shown in Fig. 1. The hydrogen diffusion apparatus is shown in Fig. 2. The steel specimen was tightened with the copper gasket. Concavity at the center of the specimen could be prevented by using the stainless steel support just

**Table I. Chemical Composition, History, and Size of the Iron Specimen**

Chemical composition, wt pct:
0.045 C, 0.39 Mn, 0.004 Si, 0.003 Al, 0.002 N, 0.007 P, 0.014 S

History and size:

A) Fully-annealed
1) History: Hot-rolled to 0.100 in., cold-rolled to 0.035 in., and then fully-annealed. Sectional grain size: ASTM No. 7.
Size: Diam = 7.6 cm, thickness = 0.035 in. (0.0888 cm).

B) Slightly cold-worked
2) History: Fully-annealed and then slightly-filed (one side).
Size: Diam = 7.6 cm, thickness = 0.025 in. (0.0635 cm).
3) History: Fully-annealed and then 5 pct cold-rolled.
Size: Diam = 7.6 cm, thickness = 0.033 in. (0.0838 cm).
4) History: Fully-annealed and then one-side machined from 0.035 to 0.015 in. (0.038 cm).

**Table II. Experimental Conditions**

1) Abrasion condition in distilled water and Buffer solution
Grinder diam = 4 cm
Grinder grain size = emery polishing paper grit 220 (corresponding)
Total weight = 300 g
Grinding speed = 20 rpm.
2) Vacuum system = $5 \times 10^{-9}$ torr.
3) Hydrogen standard leak = $1.9 \times 10^{-7}$ (STP cc $H_2$ per sec)
Sensitivity of mass spectiometer = $0.85 \times 10^{-9}$ STP cc $H_2$ per sec per division in Multiplier 30.

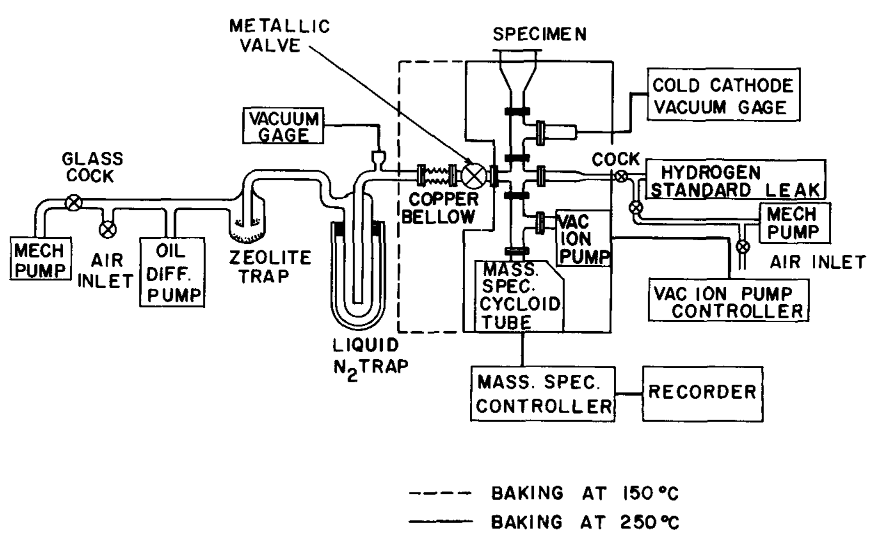


Fig. 1—Experimental apparatus for the study of hydrogen diffusion.

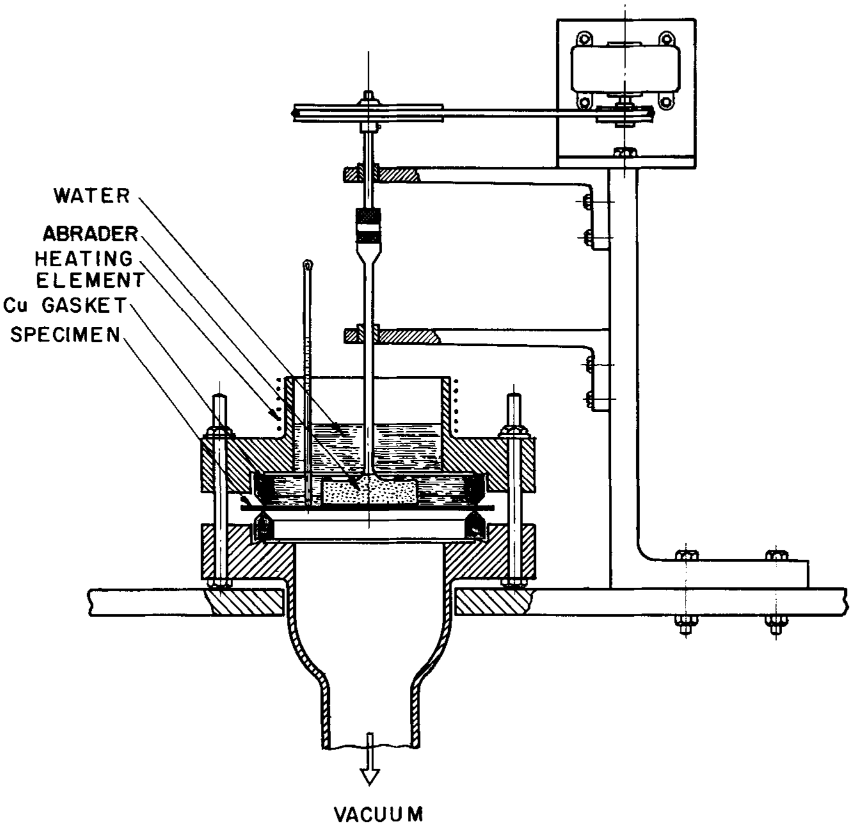


Fig. 2—Hydrogen diffusion apparatus.

below the specimen in the vacuum system, when the specimen was thin.

The author found that hydrogen could diffuse into and through a fully-annealed iron specimen from water without abrasion only when water contacted the specimen. The author could not observe this hydrogen diffusion phenomenon in a steam atmosphere. In general, hydrogen is considered to diffuse into a steel specimen only during abrasion because of the formation of an iron oxide film on the surface of the specimen contacting water. This consideration is not generally true, but is under the special experimental conditions.

The diffusion time was taken as zero when distilled water was supplied to the specimen. Keeping the specimen temperature at the desired constant temperature was easy while water existed. As soon as distilled water was removed and the specimen wiped with paper, a distinct decay profile, as shown in Fig. 3, was obtained. To prevent temperature change in the liquid-removal technique, a massive, moisture-free copper disc, which was kept at the same temperature, was brought in contact with the surface of the specimen. Fig. 3 shows a typical example of the hydrogen flux, $J$, vs diffusion time, $t$, for distilled water at 52°C in contact with a fully-annealed iron specimen containing no active traps for diffusing hydrogen. Consider first the no-abrasion case. When water contacted the iron specimen, a slow increase of hydrogen flux, $J$, was observed for $6 \times 10^3$ sec. Beyond this value, the hydrogen flux increased rapidly with time following the curve shown in Fig. 3 and arrived at the saturated value at steady state in about $10^4$ sec. As soon as distilled water was removed from the specimen, the hydrogen flux decreased rapidly following the distinct decay curve. The best fit was found between the experimental curve (solid line) and theoretical curves (broken lines) for one-sided decay and two-sided decay. As an example of calculating $D$ and $C_S$, the theoretical curve having the best fit with the experimental curve in one-sided decay using Eq. [6] has $\tau = 45$. From Eq. [3], $D = 1.8 \times 10^{-5}$ sq cm per sec. Since $(J)_\infty \simeq 9 \times 10^{-10}$ (STP cc per sq cm · sec) at steady state, one can obtain $C_S \simeq 4.5 \times 10^{-6}$ (STP cc

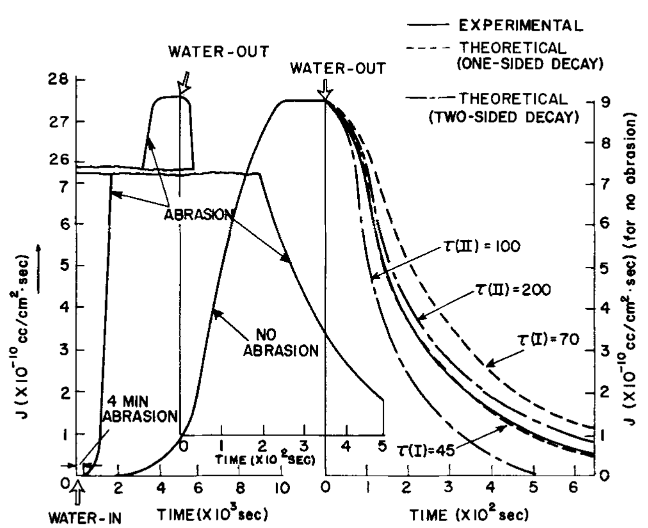


Fig. 3—Flux and time for the fully-annealed iron specimen, 0.035 in. thick, in distilled water at 52° C.

per cu cm) from Eq. [4] and $P \simeq 8 \times 10^{-11}$ (STP cc per sq cm · sec per cm thickness).

Considered statistically, one-sided decay made a better fit with the experimental curve than two-sided decay, even though it is still hard to say which mechanism applies, in fact, for hydrogen diffusion. This point will be treated more in the discussion. The value of $D$ calculated by one-sided decay was about five times that calculated by two-sided decay. Eq. [2] for the stabilization technique could not be used because of too large a difference between the theoretical curves and experimental curve. The reader should note that the time scale for the build-up curve is in $10^3$ sec units whereas that for the decay curve is in $10^2$ sec units. This would suggest that the stabilization process was, at least in part, rate-limited at the surface.

In the case of abrasion, the hydrogen flux increased more rapidly than in the case of no abrasion. Right after a 4-min abrading operation in distilled water at 52°C, the saturated value was reached in $4 \times 10^3$ sec. As soon as water was removed from the specimen, a a distinct decay profile was obtained as shown in Fig. 3.

The pH change of the Buffer solution at elevated temperatures is not important, since the hydrogen concentration at the specimen surface contacting the liquid, $C_S$, was calculated directly from the scanned maximum height on the recorder of the mass spectrometer. The recorder of the mass spectrometer showed correctly the hydrogen flux at the specimen surface at the moment of observation. Therefore, the Buffer solution was only used to give the necessary hydrogen concentration at the specimen surface for measurement of the diffusion coefficient of hydrogen.

## RESULTS

1) Results for a Fully-Annealed Iron Specimen. The calculated values for the saturated hydrogen concentration, $C_S$,* at the surface of a fully-annealed iron specimen contacting a Buffer solution ($Na_2HPO_4$-Citric Acid) with pH 6.4, and permeability, $P$, are summarized in Tables III and IV, respectively.

*The units of the hydrogen concentration at the specimen surface contacting the distilled water, $C_S$, can be expressed as follows, since one hydrogen molecule detected in the vacuum system by means of the mass spectrometer, $H_2$, comes from the combination of two hydrogen atoms in the lattice of iron specimen:

$$C_S\left(\text{in } \frac{\text{atom H}}{\text{cm}^3\text{ Fe}}\right) = 2\left(\frac{\text{atom H}}{\text{molecule H}_2}\right)\left[\frac{6.02 \times 10^{23}\ (\text{molecule H}_2/\text{mole})}{22.4 \times 10^3\ (\text{STP cc H}_2/\text{mole})}\right] \times C_S\left(\text{in } \frac{\text{STP cc H}_2}{\text{cm}^3\text{ Fe}}\right) = 0.538 \times 10^{20}\, C_S\left(\text{in } \frac{\text{STP cc H}_2}{\text{cm}^3\text{ Fe}}\right)$$

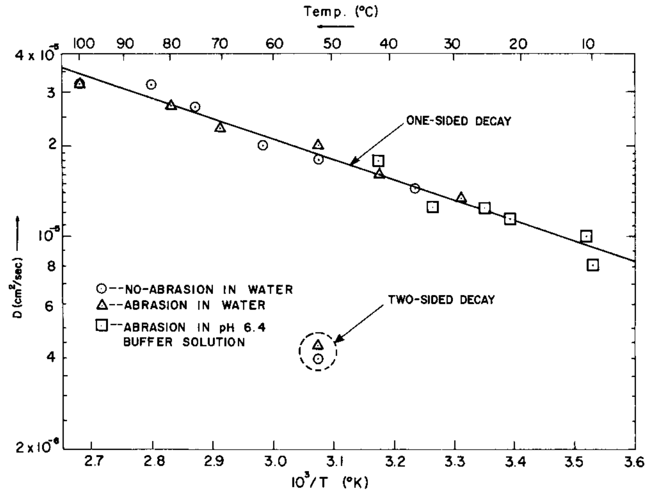


Fig. 4—Log $D$ vs $1/T$ for the fully-annealed iron specimen 0.035 in. thick.

The permeability, $P$, of hydrogen in a fully-annealed iron specimen contacting distilled water with the temperature range 36° to 100°C for the case of no abrasion is given by

$$P = 7.2 \times 10^{-2} \exp\left(-\frac{13{,}000}{RT}\right) (\text{STP cc/cm}^2 \cdot \text{sec per cm thickness})$$

The log of the hydrogen diffusion coefficient for a fully-annealed iron specimen vs $1/T$ is shown in Fig. 4.

A better fit between theoretical and experimental curves is obtained the higher the hydrogen concentration at the specimen surface contacting liquid is and the lower the background hydrogen in the vacuum system.

Since there is no difference in diffusion coefficients between the cases of no abrasion, abrasion, and pH 6.4 Buffer solution, one may say that in the case of fully-annealed iron, there is no effect of the specimen surface or of hydrogen ions upon the hydrogen diffusion coefficient, also the abrading operation and the decreased pH increase the hydrogen concentration at the surface of the specimen contacting distilled water or Buffer solution. The hydrogen diffusion coefficient for a fully-annealed iron in the temperature range 10° to 100°C according to the one-sided decay technique is

**Table III. Values of the Saturated Hydrogen Concentration at the Surface Contacting Water or pH 6.4 Buffer Solution ($Na_2HPO_4$-Citric Acid), $C_S$ (STP cc $H_2$/cm³ Fe) for the Fully-Annealed Iron Specimen 0.035 in. Thick**

| Temp, °C | | 10 | 11 | 21 | 25 | 29 | 33 | 36 | 42 | 52 | 62 | 70 | 75 | 80 | 84 | 100 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $C_S \times 10^{-6}$ | No abrasion in water | – | – | – | – | – | – | 4.7 | – | 4.5 | 12 | – | 10 | – | 25.7 | 75.5 |
| Time in water | | – | – | – | – | – | – | 31 hr | – | 2.7 hr | 2.3 hr | – | 44 min | – | 44 min | 25 min |
| $C_S \times 10^{-6}$ | Abrasion in water | – | – | – | – | 17.2 | – | – | 13.9 | 12.3 | – | 44.8 | – | 83 | – | 105 |
| Abrasion time, min | | – | – | – | – | 3.4 | – | – | 9 | 4 | – | 1.5 | – | 1 | – | 1 |
| Time in water | | – | – | – | – | 9.4 hr | – | – | 68 min | 66 min | – | 49 min | – | 27 min | – | 21 min |
| $C_S \times 10^{-3}$ | Abrasion in pH 6.4 Buffer solution | 1.1 | 1.1 | 1.3 | 2.5 | – | 4.0 | – | 4.4 | – | – | – | – | – | – | – |
| Abrasion time, min | | 2 | 2 | 2 | 2 | – | 1 | – | 1 | – | – | – | – | – | – | – |

$$D = 2.2 \times 10^{-3} \exp\left(-\frac{3100}{RT}\right) \text{ cm}^2/\text{sec}$$

From the equation above, $Q = 3.1$ kcal per mole, $D_0 = 2 \times 10^{-3}$ sq cm per sec. The entropy term, $D_0$, in the interstitial mechanism will be shown by $D_0 = \alpha a_0^2 \nu \exp(\Delta S/R)$. Simply taking the jump distance of hydrogen atom, $a_0 \simeq 10^{-8}$ cm, the geometry factor in bcc, $\alpha = \frac{1}{6}$ and assuming the jump frequency of hydrogen atom, $\nu \simeq 10^{13}$ (1 per sec), $\Delta S/R \simeq 2.5$ or $\Delta S \simeq 5$ (cal per °K · mole). The value of $\Delta S$ seems to be reasonable considering those for carbon and nitrogen.[12,13]

2) Results of Trapping Phenomenon for Slightly Cold-Worked Iron. In order to study the trapping phenomenon of hydrogen atoms in cold-worked iron in contact with water or acidic Buffer solution, the same experimental technique was applied to the following three kinds of slightly cold-worked iron specimens: a) Slightly-filed (one side) specimen of fully-annealed iron, 0.025 in. thick. b) 5 pct cold-rolled iron specimen, 0.033 in. thick. c) One-side-machined iron specimen (final thickness = 0.015 in.). A fully-annealed iron specimen was machined from 0.035 to 0.015 in. in thickness.

The annealed surface of the specimen a) or c) was brought in contact with the liquid. The apparent diffusion coefficients for the above three kinds of slightly cold-worked specimens with the same chemical composition shown in Table I are as follows:

For the iron specimen with one-side filed slightly, 0.025 in. thick,

$$D_{\text{app}} = 8.6 \times 10^{-2} \exp\left(-\frac{5900}{RT}\right) \text{ cm}^2/\text{sec}$$

For the 5 pct cold-rolled iron specimen, 0.033 in. thick,

$$D_{\text{app}} = 0.31 \exp\left(-\frac{6900}{RT}\right) \text{ cm}^2/\text{sec}$$

For the one-side-machined iron specimen, 0.015 in. thick,

$$D_{\text{app}} = 1.2 \exp\left(-\frac{8000}{RT}\right) \text{ cm}^2/\text{sec}$$

The plot of log $D$ vs $1/T$ of the above results for the three kinds of slightly cold-worked specimens is shown with the result of the fully-annealed iron, 0.035 in. thick, in Fig. 5.

From the results of the ratio, $(k/p)/V_0 \simeq 10^5$ for the one-side-machined iron specimens as shown in Table V, it appears that the size of the active trap is

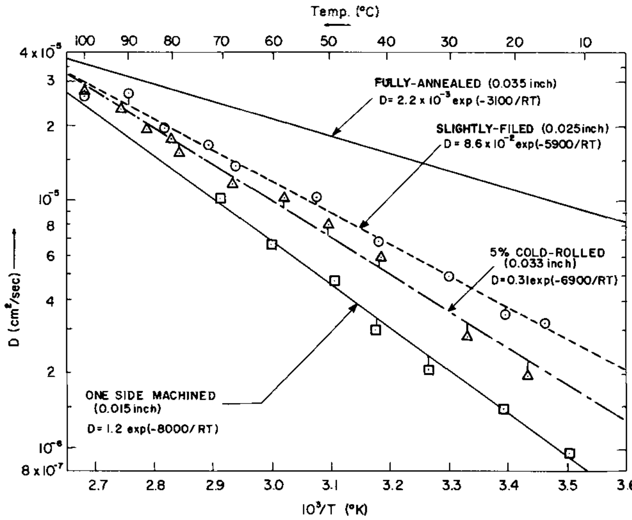


Fig. 5—Log $D$ vs $1/T$ for slightly-cold-worked iron specimens in comparison with the fully-annealed specimen.

Table IV. Permeability, $P$ (STP cc/cm² · sec per cm thickness) for the Same Experimental Conditions as in Table III

| Temp, °C | | 10 | 11 | 21 | 25 | 29 | 33 | 36 | 42 | 52 | 62 | 70 | 75 | 80 | 84 | 100 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Distilled water | No abrasion, × $10^{-11}$ | – | – | – | – | – | – | 6.8 | – | 8.0 | 23.4 | – | 27 | – | 82 | 242 |
| | Abrasion, × $10^{-11}$ | – | – | – | – | 23 | | – | 22 | 25 | – | 103 | – | 224 | – | 336 |
| pH 6.4 Buffer Solution | Abrasion, × $10^{-8}$ | 0.9 | 1.1 | 1.5 | 3.1 | – | 4.9 | – | 7.9 | – | – | – | – | – | – | – |

Table V. Trapping Parameters for the One-Side-Machined Iron Specimen 0.015 in. Thick

| Temp, °C | 12 | 22 | 33 | 42 | 49 | 60 | 70 |
|---|---|---|---|---|---|---|---|
| $D$, sq cm per sec, × $10^{-6}$ | 8.8 | 11.3 | 13.8 | 15.8 | 17.6 | 20.8 | 23.8 |
| $D_{\text{app}}$, sq cm per sec, × $10^{-6}$ | 0.92 | 1.47 | 2.1 | 2.94 | 4.7 | 6.7 | 10 |
| $C_s$, cc $H_2$ per cu cm Fe, × $10^{-4}$ | 3.9 | 5.3 | 5.5 | 8.1 | 2.7 | 1.26 | 1.34 |
| $C_s$, atom H per cu cm Fe, × $10^{15}$ | 21 | 28.4 | 29.8 | 43.6 | 14.4 | 6.8 | 7.2 |
| $P$, cc $H_2$ per sq cm · sec, × $10^{-10}$ | 3.6 | 7.8 | 11.5 | 23.8 | 12.7 | 8.4 | 13.4 |
| $Q_e/Q_a$ | 0.82 | 0.73 | 0.54 | 0.36 | 0.32 | 0.86 | 0.62 |
| $k/p$, cu cm Fe per atom H, × $10^{-16}$ | 0.42 | 0.35 | 0.4 | 0.3 | 0.94 | 1.23 | 1.54 |
| $C_s k/p$ | 0.89 | 1.0 | 1.18 | 1.32 | 1.35 | 0.84 | 1.11 |
| $Nk/p$ | 8.58 | 6.68 | 5.58 | 4.37 | 2.75 | 2.12 | 1.38 |
| $N$, trap per cu cm Fe, × $10^{16}$ | 20 | 19.1 | 14.1 | 14.4 | 2.92 | 1.72 | 0.9 |
| $N/C_s$ | 9.6 | 6.7 | 4.7 | 3.3 | 2.03 | 2.54 | 1.25 |
| $k/p/V_0$, × $10^5$ | 1.2 | 1 | 1 | 0.9 | 2.8 | 3.6 | 4.5 |
| Abrasion time | 3 min | 2 min | 2 min | 2 min | 1 min | 2 min | 1 min |
| Liquid | pH 6.4 B.S. | pH 6.4 B.S. | pH 6.4 B.S. | pH 6.4 B.S. | pH 6.4 B.S. | pH 6.8 B.S. | pH 6.8 B.S. |

considerably larger than a single lattice atom spacing as usually considered. The calculated value of the energy depth of the active trap, $Q_t$, from Eq. [9] using $V_0 = 3.4 \times 10^{-22}$ cu cm and $(k/p)/V_0$ value at room temperature is 3.4 kcal per mole for the one-side-machined iron specimen, 0.015 in. thick. From here, the total energy barrier for hydrogen atom jumping out of an active trap will be 6.5 kcal per mole (=3.1 + 3.4), assuming that active traps have the same energy depth of 3.4 kcal per mole. Since the apparent activation energy measured for the one-side-machined specimen is 8 kcal per mole, the total energy well of the active trap should be bigger than 8 kcal per mole, when we consider the weighting factor[14] for the activation energies of hydrogen atoms which diffuse through the active traps and the normal lattice containing no active traps. If hydrogen atoms are captured at dislocations, the energy well of an active trap, $Q_t$, will be 6.4 kcal per mole, *i.e.*, the average hydrogen-dislocation binding energy.[15] Therefore, the mean total energy barrier for a hydrogen atom jumping out of an active trap (dislocation) will be 9.5 kcal per mole (=6.4 + 3.1). Since the apparent activation energy is smaller than 9.5 kcal per mole, one may assume that not all of the hydrogen in this specimen is trapped by active traps, *i.e.*, dislocations. It seems that the calculated value of the energy well of active traps, $Q_t$, from the trapping parameters is inconsistent with Eq. [9]. This inconsistency seems to come from using the incomplete treatment[16] of the mathematical equations and inaccuracies in the calculated trapping parameters arising from uncertainties in the mathematical solutions to the trapping theory.

## DISCUSSION

The liquid-removal technique is a good method for determining the hydrogen diffusion coefficient in metals and gives reproducible data.

The summary of literature data for diffusion coefficients of hydrogen in stress-free $\alpha$-Fe including this study is shown in Table VI and Fig. 6. The fine solid lines in Fig. 6 are for pure iron, while the fine broken lines are for mild steel or low alloy steel. The heavy solid line is the result obtained by one-sided decay, while the heavy broken line is the result obtained by two-sided decay in this study. One can see that the result obtained by one-sided decay coincides very well with the results by other investigators.[3,6,9,17-23] It is interesting that the result due to one-sided decay coincides much better with the data by other investigators than that due to two-sided decay. From the result that one-sided decay made a better fit statistically with the experimental curve than two-sided decay, it will be likely that a larger amount of hydrogen diffused toward the vacuum side of the specimen than the side exposed to the atmosphere, even though it is still hard to say which mechanism was operative.

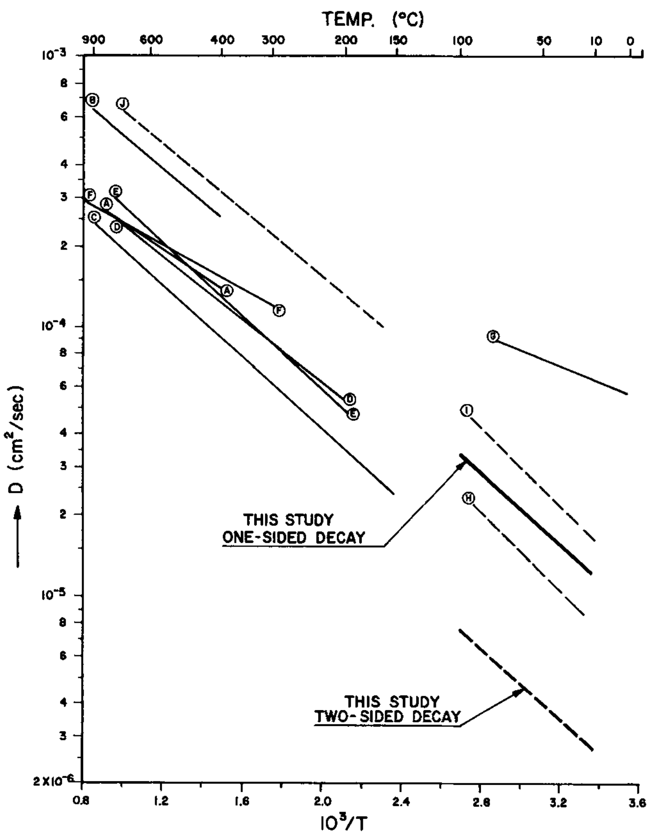


Fig. 6—Log $D$ vs $1/T$ for the summary of literature data including this study as shown in Table VI for diffusion coefficient of hydrogen in stress-free $\alpha$-Fe.

**Table VI. Summary of Literature Data for Diffusion Coefficient of Hydrogen in Stress-Free $\alpha$-Fe**

| Investigator and Ref. | Material | Method | Temp Range, °C | $D_0$, $\times 10^{-3}$, sq cm per sec | $Q$, kcal per mole | Mark in Fig. 6 |
|---|---|---|---|---|---|---|
| Sykes, Burton, and Gegg[17] | Iron | Permeability | 400° to 900° | 0.76 | 2.3 | A |
| Geller and Sun[18] | Iron | Permeability | 400° to 900° | 2.2 | 2.9 | B |
| Stross and Tompkins[19] | Iron | Evolution | 150° to 900° | 5.3 | 3.05 | C |
| Eichenauer, Künzig, and Pebler[20] | Iron | Evolution | 200° to 774° | 0.93 | 2.7 | D |
| Johnson and Hill[6] | Iron | Evolution | 200° to 780° | 1.4 | 3.2 | E |
| Heumann and Primas[21] | Iron | Evolution | 300° to 1100° | 0.64 | 1.92 | F |
| Beck, Bockris, McBreen, and Nanis[22] | Iron | Permeability (Electrochem) | 10° to 75° | 0.6 | 1.33 | G |
| Smialowski[23] | Mild steel | Time lag | 30° to 90° | 2.1 | 3.3 | H |
| Frank, Swets, and Fry[3] | Mild steel | Build-up | 26° to 90° | 5 | 3.4 | I |
| Nelson[9] | 0.7 pct Cr-0.2 pct Mo steel | Permeability | 170° to 700° | 2.6 | 2.8 | J |
| Present study | Iron | One-sided decay (Liquid-removal) | 10° to 100° | 2.2 | 3.1 | |

From the results of log $D$ vs $1/T$ shown in Fig. 5 for specimens with 0.025 and 0.015-in. thickness, respectively, in which the one side was fully-annealed and the other side was cold-worked, the diffusion coefficient of hydrogen tends to decrease with the thickness of the specimen because of the effect of cold-working. This result coincides with those obtained by the other workers,[24] in which the diffusion coefficient of hydrogen decreased with the thickness ($\lesssim$0.08 cm) and was constant with the thickness (>0.08 cm) of the specimen.

From the results of fully-annealed specimens thicker than 0.08 cm as shown in Figs. 4 and 6, one may see there is no dependence of $D$ on thickness. There appears to be no effect of vacuum-induced deformation on the value of $D$ for specimens thicker than 0.08 cm, in that the same values of $D$ occurred in the case of no stainless steel support just below the specimen as were obtained in the case of using a support.

A question arises whether or not $D$ is constant with decreasing thickness in fully-annealed, unstrained specimens thinner than 0.08 cm. This question could be clarified experimentally if one could establish whether stress level has any influence. For such an experiment, a method as accurate as mass spectrometry would be required, but without direct application of vacuum on the specimen, since this gives a higher stress the thinner the specimen.

It seems that the decrease of $D$ with the thickness of the thin specimen ($\lesssim$0.08 cm) results from strain induced in the specimen during handling and from deflection during the experiment. In the ideal case of an unstrained, fully-annealed sample, it is possible that $D$ is independent of thickness.

The permeability of hydrogen, $P$, will be changed by the conditions of the specimen and experiment for the following reasons.

1) Effect of $C_S$: The permeability of hydrogen, $P$, will change when the value of $C_S$ is not constant. The addition of acidic Buffer solution to water and the light abrading operation* in liquid increase the values of $C_S$ and permeability.

2) Effect of cold working: The traps in the bulk of the specimen created by cold working will decrease the permeability of hydrogen by trapping the diffusing hydrogen atoms.

The value of permeability of hydrogen obtained in this study will be possibly more accurate than that obtained from the conventional method since even extremely small flux of hydrogen could be detected by the mass spectrometer. As far as the author knows, no other data resulting from the liquid-removal method by mass spectrometer is available for the permeability of hydrogen in fully-annealed (stress-free) pure iron with no-abrasion in contact with distilled water.

For these reasons, it is meaningless to compare the values of activation energies of the permeability of hydrogen, which were obtained using different conditions of the specimen and experimental method, except for appropriate results, since the permeability is able to change relatively widely.

*Light abrasion in liquid increases the values of $C_S$ and permeability as shown in Fig. 3. It seems that the permeability of hydrogen is increased by the surface abrasion which is so light that negligible number of active or deep traps for diffusing hydrogen atoms are created in the bulk.

## CONCLUSION

1) Hydrogen can diffuse into iron only when the iron contacts water particularly in the case of fully-annealed iron.

2) Hydrogen permeability increases with temperature and is accelerated by abrasion in liquid and by hydrogen ions in dilute acid because of the increase of the hydrogen atom concentration at the iron surface contacting water or dilute acid.

3) Permeability of hydrogen for fully-annealed iron contacting distilled water without abrasion in the temperature range 36° to 100°C is

$$P = 7.2 \times 10^{-2} \exp\left(-\frac{13{,}000}{RT}\right) \text{ STP cc } H_2/\text{cm}^2\cdot\text{sec/cm thickness of iron}$$

4) The diffusion coefficient of hydrogen for fully-annealed iron in the temperature range 10° to 100°C is

$$D = 2.2 \times 10^{-3} \exp\left(-\frac{3100}{RT}\right) \text{cm}^2/\text{sec}$$

5) Cold-working results in lower values of $D$ and higher apparent activation energies.

6) The diffusion coefficient of hydrogen seems to decrease with the thickness in the thin iron specimen, because the thinner specimen may be more affected by cold-working.

7) The size of the active trap, $k/p$, the trap density, $N$, and the ratio between the size of the active trap and the space volume of a normal unit lattice containing one interstitial hydrogen atom, $(k/p)V_0$, were calculated.

## ACKNOWLEDGMENTS

The author is grateful to Prof. G. M. Pound, Prof. H. W. Paxton, and staff members of Raychem Corp. for useful discussion and comments.

The author wishes to express profound thanks to Prof. P. G. Shewmon and Prof. H. W. Paxton for the kind arrangements and support of this study so the author might come from Korea and work at Carnegie-Mellon University. Many thanks are due to Dr. J. D. Harrison, Mr. R. L. Caesar, and Mr. K. L. Wassam for the review of English of the manuscript. The author wishes to thank Prof. R. H. Lambert, Dr. J. W. Newsome, Dr. R. Judd, Mr. W. Poling, and Mr. G. G. Biddle for providing the specimens and some parts of the experimental apparatus. This research was supported by the Advanced Research Projects Agency of the Department of Defense under contract NONR-760(31).

## REFERENCES

1. F. J. Norton: *J. Appl. Phys.*, 1940, vol. 11, p. 262.
2. L. S. Darken and R. P. Smith: *Corrosion*, 1949, vol. 5, p. 1.
3. R. C. Frank, D. E. Swets, and D. L. Fry: *J. Appl. Phys.*, 1958, vol. 29, p. 892.
4. D. E. Swets, R. C. Frank, and D. L. Fry: *Trans. TMS-AIME*, 1958, vol. 212, p. 219.
5. M. L. Hill and E. W. Johnson: *Trans. TMS-AIME*, 1959, vol. 215, p. 717.
6. E. W. Johnson and M. L. Hill: *Trans. TMS-AIME*, 1960, vol. 218, p. 1104.
7. A. McNabb and P. K. Foster: *Trans. TMS-AIME*, 1963, vol. 227, p. 618.
8. P. K. Foster, A. McNabb, and C. M. Payne: *Trans. TMS-AIME*, 1965, vol. 223, p. 1022.

9. H. G. Nelson: NASA-Ames Research Center, Field, Calif., unpublished research, 1967.
10. J. Crank: *The Mathematics of Diffusion,* p. 45, Oxford University Press, N. Y., 1956.
11. R. A. Oriani: *Proc. Conf. Fundamental Aspects of Stress Corrosion Cracking,* 1967, p. 32, National Association of Corrosion Engineers, 1969.
12. C. Wert: *Phys. Rev.,* 1950, vol. 79, p. 601.
13. C. Wert: *J. Appl. Phys.,* 1950, vol. 21, p. 1196.
14. N. A. Gjostein: *Metals Surfaces,* ASM-AIME Seminar Book, 1963, p. 99.
15. R. Gibala: *Trans. TMS-AIME,* 1967, vol. 239, p. 1574.
16. R. A. Oriani: Research Center, U. S. Steel Corp., Monroeville, Pa., private communication.
17. C. Sykes, H. H. Burton, and C. C. Gegg: *J. Iron Steel Inst.,* 1947, vol. 156, p. 155.
18. W. Geller and T. H. Sun: *Arch. Eisenhuettenw.,* 1950, vol. 21, p. 423.
19. T. M. Stross and F. C. Tompkins: *J. Chem. Soc. London,* 1956, Part 1, p. 230.
20. W. Eichenauer, H. Künzig, and A. Pebler: *Z. Metallk.,* 1958, vol. 49, p. 220.
21. T. Heumann and D. Primas: *Z. Naturforsch,* 1966, vol. 21A, p. 260.
22. W. Bech, J. O'M. Bockris, J. McBreen, and L. Nanis: *Proc. Roy. Soc.,* 1966, vol. A290, p. 220.
23. A. Smialowski: *Neue Huette,* 1957, vol. 2, p. 621.
24. E. E. Fletcher and A. R. Elsea: Hydrogen Movement in Steel, Defense Metals Information Center, Battelle Memorial Institute, 1965, p. 51.